Karaayvas 2018 TNBC — analysis summary

RUN 1 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony(θ=5) integrates all 5 patients: same-sample neighbour fraction 0.82 → 0.28 (null 0.25), iLISI median 1.97 → 4.12 of 5. One connected component (0 islands). Judged on the regenerated embedding + per-neighbourhood iLISI, not a scalar: samples intermix AND the manifold is continuous.

Gate 2 — module geography → GO (non-branching)

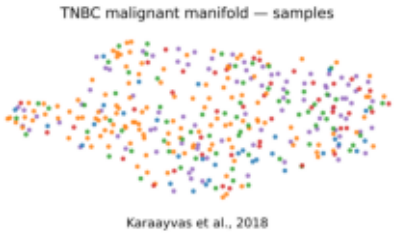
Non-branching case: a single pSom-origin → terminus developmental gradient. pSom falls along pseudotime (ρ −0.41); Gast & Crest both rise to the terminal tip (ρ +0.79 / +0.82; Crest decile pt=0.98). Joint pSom&Crest enrichment 0.37 (<1 = inverse terminal gradient, not a co-expression hub). FLAG (pass anyway): Crest co-enriches the Gast pole (smoothed Gast↔Crest r=+0.59), an inversion of the expected pSom-arm Crest — Gast, not pSom, tracks the neural-crest terminus here.

Gate 3 — hub cross-repression → NA

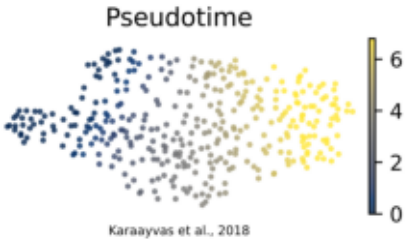
Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks, fold-free). No bifurcation, so the branching-only hub cross-repression test is not run (correlation / modules notebooks not produced, per spec).

STEP FIGURES

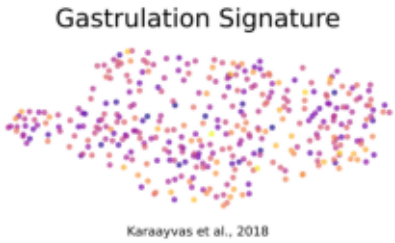
Manifold — samples



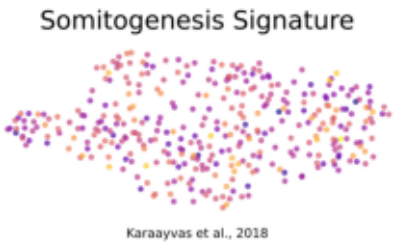
Pseudotime



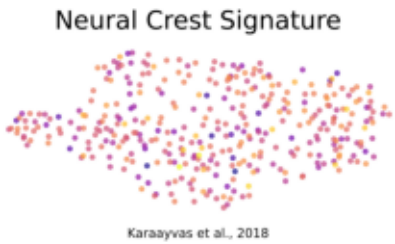
Gastrulation



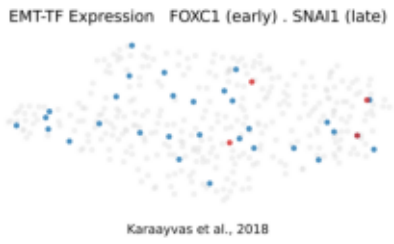
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Triple-negative breast cancer (Karaayvas et al., 2018), SmartSeq2 — 360 malignant cells over 5 patients; single low-cycling body (~14% cycling), cell-cycle-free axis (pseudotime vs cycling ρ −0.19).

Verdict.
Pattern reproduces in the non-branching form (Linear-pSom/Crest): one continuous somitogenesis-origin → neural-crest terminus axis. Gate 1 GO / Gate 2 GO / Gate 3 NA, with the Crest-arm inversion flagged (Crest tracks the Gast terminus). EMT-TFs FOXC1 (early/root) → SNAI1 (late).

Parameter choices

Sample threshold	≥12 (geom-mean of 31→5 cliff, ratio 6.2) → 5/6 samples, 360 cells	**Overdispersed genes**	51 (scFates find_overdispersed; sparse SmartSeq2)
Integration	Harmony on sample, θ=5 (mixing near null then flattens)	**Embedding**	n_pcs=20, n_neighbors=30, spread=1.3; 1 component, 0 islands removed
Geometry	Non-branching scFates curve N=20 λ=0.3 μ=100; 2 tips/0 forks, SA 2.91; corr(x,t)=0.97	**Root**	tip 2 = Crest-low, pSom-high origin
Module means (raw)	Gast −0.220 · pSom +0.106 · Crest −0.067 (Linear-pSom/Crest)	**EMT-TFs shown**	FOXC1 (early/root, 33%) ↔ SNAI1 (late/terminus); detection-gated